COHOMOLOGY OF THE STRUCTURE SHEAF IN A NODAL FAMILY WITH CONNECTED GEOMETRIC FIBRES: PUSHFORWARDS, LOCAL FREENESS AND BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the pushforward identification, the perfect decomposition with tor amplitude, the fibre cohomology computation with its vanishing, and the finite-local-freeness conclusion with base-change compatibility are all retained verbatim.

Editorial note. The label lemma-genus-in-nodal-family-of-curves names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Pushforward for proper flat morphisms with geometrically reduced connected fibres. Derived Categories of Schemes, Lemma perfect-lemma-proper-flat-geom-red-connected. Gives part (a): f_*O_X = O_S compatible with base change.
- Perfect decomposition of the total pushforward. Derived Categories of Schemes, Lemmas perfect-lemma-proper-flat-h0 and perfect-lemma-flat-proper-perfect-direct-image-general. Writes Rf_*O_X = O_S (+) P locally with P perfect of tor amplitude in [1, infinity), and gives base-change compatibility of Rf_*O_X.
- Vanishing and perfect lifting. Cohomology, Proposition cohomology-proposition-vanishing-Noetherian; More on Algebra, Lemma more-algebra-lemma-lift-perfect-from-residue-field. Kills the fibre cohomology in degrees at least 2 on the one-dimensional Noetherian fibres and promotes H^1(P) to a finite locally free module from its residue-field fibres.

Source reference. Original proposition 02UZ: sources/stacks-chow/tags.txt, source lines 3750–3750. Prerequisite pointer invoked at proof line(s) 5268; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BCD: sources/stacks-chow/tags.txt, source lines 14276–14276. Prerequisite pointer invoked at proof line(s) 5271; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0B91: sources/stacks-chow/tags.txt, source lines 14159–14159. Prerequisite pointer invoked at proof line(s) 5259; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0E0L: sources/stacks-chow/tags.txt, source lines 17541–17541. Prerequisite pointer invoked at proof line(s) 5251; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0E62: sources/stacks-chow/tags.txt, source lines 17732–17732. Prerequisite pointer invoked at proof line(s) 5253; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Assume*

(1) *f is proper,*
(2) *f is at-worst-nodal of relative dimension 1, and*
(3) *the geometric fibres of f are connected.*

Then (a) $f_*\mathcal{O}_X = \mathcal{O}_S$ and this holds after any base change, (b) $R^1 f_*\mathcal{O}_X$ is a finite locally free $\mathcal{O}_S$-module whose formation commutes with any base change, and (c) $R^q f_*\mathcal{O}_X = 0$ for $q \geq 2$.

Proof. Part (a) follows from Derived Categories of Schemes, Lemma 0E0L. By Derived Categories of Schemes, Lemma 0E62 locally on S we can write $Rf_*\mathcal{O}_X = \mathcal{O}_S \oplus P$ where P is perfect of tor amplitude in $[1, \infty)$. Recall that formation of $Rf_*\mathcal{O}_X$ commutes with arbitrary base change (Derived Categories of Schemes, Lemma 0B91). Thus for $s \in S$ we have

$$H^i(P \otimes^{\mathbf{L}}_{\mathcal{O}_S} \kappa(s)) = H^i(X_s, \mathcal{O}_{X_s}) \text{ for } i \geq 1$$

This is zero unless $i = 1$ since X_s is a 1-dimensional Noetherian scheme, see Cohomology, Proposition 02UZ. Then $P = H^1(P)[-1]$ and $H^1(P)$ is finite locally free for example by More on Algebra, Lemma 0BCD. Since everything is compatible with base change we conclude. □